Supplementary material: Bounded Instance-Support Evaluation for Finite Crop–Weed Review Queues

Shuhao Liu[a,†], Zhuo Chen[b,†,*], Zhi Ling[c], Yu Yan[c], Jie Liu[a], Qiuxue Wu[a], Ziyi Kuang[c], Haiyou Zhang[c]

[a] *Harbin University of Science and Technology, No. 52 Xuefu Road, Nangang District, Harbin 150080, Heilongjiang, China*
[b] *Chalmers University of Technology, SE-412 96 Gothenburg, Sweden*
[c] *Kiwiar Co., Ltd., Singapore*

S1. Data roles and analysis sequence

WeedsGalore and CWFID have different experimental roles. WeedsGalore supports model development, internal comparison, and uncertainty analysis on its official train, validation, and test split. CWFID supports a pre-specified cross-dataset transfer. Table S1 summarises the analysis sequence. Test truth was unavailable to queue-forming stages and entered only offline scoring.

The CWFID model list, preprocessing, component definition, fixed $K = 20$, ranking rule, and endpoints were specified before external scoring. The continuous analysis used the same images and validation-selected checkpoints to characterise score distributions; it introduced no external fitting or model selection.

S2. Public-data lineage and denominators

All reported imagery is public. SugarBeets2016 supplied source-specialist masks. WeedsGalore supplied internal development data. CWFID supplied

*Corresponding author.
Email addresses: 2420610137@stu.hrbust.edu.cn (Shuhao Liu), zhuoc@chalmers.se (Zhuo Chen), zhi.ling@kiwiar.com (Zhi Ling), yu.yan@kiwiar.com (Yu Yan), liujie@hrbust.edu.cn (Jie Liu), 2420610140@stu.hrbust.edu.cn (Qiuxue Wu), ziyi.kuang@kiwiar.com (Ziyi Kuang), haiyou.zhang@kiwiar.com (Haiyou Zhang)
†These authors contributed equally to this work.

Table 1: Data roles and analysis sequence. Recorded electronic times use UTC+08:00.

Event	Recorded time	Analysis class	Truth access
WeedsGalore U-Net protocol	Initial analysis	Official-split model evaluation	Train for fitting; validation for selection; test for scoring
Target-semantic seed runs	10–12 Aug 2026	Seed-stability analysis	Validation-selected checkpoints; test scoring only
CWFID transfer specification	12 Aug 2026, 18:46:11	Cross-dataset design	Public metadata only
Strong-model configuration	12 Aug 2026, 18:58:00	Model-family comparison	WeedsGalore validation selection; no CWFID access
Matched-burden configuration	12 Aug 2026, 19:08–19:17	Proposal comparison	Existing test outputs; no fitting on test truth
CWFID transfer execution	12 Aug 2026, 19:54:27	External evaluation	Annotations used after model outputs were fixed
Operator review	12 Aug 2026	Independent operator evaluation	RGB views only; mask truth and scores hidden
Contract, merge, and uncertainty analyses	13 Aug 2026	Sensitivity and uncertainty analysis	Immutable predictions and offline matches
CWFID continuous-score analysis	13 Aug 2026	External score diagnostic	Fixed checkpoints; no external selection

the cross-dataset transfer set. Public images and public weights remain at upstream repositories. The release records source links, access dates, licences, versions, byte counts, and hashes.

The WeedsGalore release describes one maize field, 48 recorded locations, and an 8/2/2 spatial allocation. It provides capture indices but no verified mapping to independent field units. Capture indices were not treated as farms. Across all 156 rasters, enumeration yielded 2,168 crop and 10,029 weed instances. Semantic and instance rasters agreed on role labels for every enumerated instance.

CWFID was fixed at upstream commit `176612f3506c0ce70d85518cb82ddd938a6206c8`. All 242 expected blobs passed byte-count and Git-blob SHA-1

Table 2: Official WeedsGalore split by acquisition date. Instance counts were enumerated from released rasters.

Date	Train tiles	Validation tiles	Test tiles	Test crop	Test weed
25 May 2023	32	8	8	122	392
30 May 2023	32	8	8	55	440
6 June 2023	32	8	8	68	726
15 June 2023	8	2	2	11	154
Total	104	26	26	256	1,712

checks. Sixty RGB–annotation pairs passed pairing, dimension, and colour-semantics audits. The frozen transformation produced 331 weed components of at least 16 pixels. These connected components are evaluation proxies rather than persistent plant identifiers.

S3. Supervision ledger and model construction

The source foreground and role specialists were ResNet-18 U-Net models initialised from ImageNet-1K weights. Both used ImageNet normalisation, AdamW, bfloat16 autocasting, gradient clipping at 5, and deterministic 128×128 patches. The foreground model used 32,768 patches, binary cross-entropy with positive weight 30 plus soft Dice, and 10 epochs. The role model used 16,384 role-balanced patches, weighted cross-entropy with weights $(0.2, 1, 1)$ plus plant-role Dice, and 8 epochs.

Semantic models trained on random 512×512 crops and were evaluated at 600×600. Weight decay was 10^{-4}. Candidate thresholds were 0.30, 0.50, and 0.70. Component areas were 16, 32, and 64 pixels. Mask R-CNN used a 600-pixel minimum and maximum side, mask threshold 0.50, detection-score floor 0.05, momentum 0.9, and weight decay 5×10^{-4}.

S4. Matching contracts, proposal selection, and crop exposure

For truth instance I, candidate C, all labelled plant pixels L, and pixels of role R, the offline scorer records

$$q_I = \frac{|I \cap C|}{|I|}, \qquad q_L = \frac{|C \cap L|}{|C|}, \qquad q_R = \frac{|C \cap R|}{|C \cap L|}.$$

Table 3: Complete supervision ledger. "Selection access" identifies labels used to choose an operating point.

Information	Available amount	Selection access	Purpose
SugarBeets pixel masks	7,050 train; 901 validation	Source fitting and diagnostics	Foreground and role specialists
WeedsGalore training masks	104 tiles	Proposal commissioning and model fitting	Dense loss, components, candidate labels
WeedsGalore validation masks	26 tiles	Epoch, threshold, area, regularisation, rank policy	Model and queue selection
WeedsGalore test masks	26 tiles	None	Offline proposal, matching, IoU, and exposure scores
Restricted candidate draw	300 training candidates per draw	All 2,436 validation candidates retained	Restricted training reveal, not a 300-label pipeline
CWFID annotations	60 images	None	One locked score and post-lock diagnostics
Operator responses	518 regions per operator	None	Agreement and human–mask comparison

Table 4: Target-trained model recipes. Public initialisation weights were stored by hash before training.

Model	Initialisation	Loss	Training	Validation selection
ResNet-18 U-Net	ImageNet-1K	weighted CE + crop/weed Dice	AdamW, 3×10^{-4}; 40 epochs; 5 seeds	epoch, threshold, area, rank score
DeepLabV3+–ResNet-50	ImageNet-1K	weighted CE + crop/weed Dice	AdamW, 3×10^{-4}; 30 epochs; 3 seeds	epoch, threshold, area
Mask R-CNN–ResNet-50-FPN	COCO V1	standard detection and mask losses	SGD, 5×10^{-3}; 20 epochs; 3 seeds	epoch

The primary role-qualified edge requires $q_I \geq 0.50$, $q_L \geq 0.50$, $q_R \geq 0.90$, and a role match. The 27-contract grid crosses $q_I \in \{0.25, 0.50, 0.75\}$, $q_L \in \{0.25, 0.50, 0.75\}$, and $q_R \in \{0.50, 0.75, 0.90\}$.

Table 5: Proposal operating-point grid on the official training split. Feasible settings have at most 100 candidates per tile on average.

Threshold / area (px)	Candidates/tile	Spatial recall	Qualified recall	Feasible
0.30 / 8	281.35	0.7135	0.2406	No
0.30 / 32	142.04	0.7000	0.2297	No
0.50 / 8	259.49	0.6325	0.2822	No
0.50 / 32	123.52	0.6190	0.2707	No
0.70 / 8	207.48	0.5416	0.2979	No
0.70 / 32	95.07	0.5271	0.2856	Yes
0.80 / 8	168.63	0.4806	0.2959	No
0.80 / 32	76.99	0.4672	0.2835	Yes
0.90 / 8	115.50	0.3703	0.2421	No
0.90 / 32	52.38	0.3587	0.2317	Yes
0.95 / 8	75.71	0.2452	0.1685	Yes
0.95 / 32	36.83	0.2366	0.1605	Yes

The deterministic selection order was higher training qualified recall, lower mean burden, higher threshold, and larger area. The selected setting was 0.70/32. The original setting was 0.95/32. The burden cap of 100 limited candidate extraction and feature computation. The downstream $K = 20$ limit represented a separate review-capacity experiment.

Table 6: Split and merge sensitivity across 156 WeedsGalore tiles. A merge overlaps at least two instances at the listed coverage threshold.

Setting	Instance coverage	Split instances	Merge candidates
Original	0.10	224	735
Original	0.25	40	640
Original	0.50	0	473
Commissioned	0.10	154	1,136
Commissioned	0.25	26	1,065
Commissioned	0.50	0	938

Table 7: Crop-exposed tile frequency for the three target-trained model families at $K = 20$. Values are mean $\pm$ sample SD across seeds.

Model	Any overlap	Crop fraction ≥ 0.01	Crop fraction ≥ 0.10
U-Net	0.2462 ± 0.0439	0.2462 ± 0.0439	0.2385 ± 0.0501
DeepLabV3+	0.2436 ± 0.0222	0.2436 ± 0.0222	0.2308 ± 0.0000
Mask R-CNN	0.2308 ± 0.0769	0.2051 ± 0.0588	0.1538 ± 0.0666

S5. Ranker-source and commissioning sensitivity

Four ranker-training schemes were evaluated without changing the 471-candidate comparison burden. Commissioned-only and original-only used candidate labels from one proposal distribution. Union training combined both distributions. Reciprocal cross-pool training applied each distribution's ranker to the other distribution.

Table 8: Primary-contract metrics under each ranker-training scheme. Ceiling and oracle values do not depend on ranker training.

Ranker	Setting	Precision	Spatial	Set	One-to-one	Set ceiling	$K = 20$ oracle
Commissioned	Original	0.5860	0.1875	0.1197	0.0970	0.1928	0.1396
Commissioned	Commissioned	0.5180	0.2693	0.1571	0.1157	0.3026	0.1618
Original	Original	0.5881	0.1682	0.1046	0.0859	0.1928	0.1396
Original	Commissioned	0.2378	0.0415	0.0350	0.0345	0.3026	0.1618
Union	Original	0.5924	0.1893	0.1221	0.0981	0.1928	0.1396
Union	Commissioned	0.5223	0.2739	0.1600	0.1192	0.3026	0.1618
Reciprocal	Original	0.5860	0.1875	0.1197	0.0970	0.1928	0.1396
Reciprocal	Commissioned	0.2378	0.0415	0.0350	0.0345	0.3026	0.1618

Table 9: Direction of commissioned-minus-original effects across 27 matching contracts. Entries are positive/zero/negative contracts.

Ranker	Precision	Spatial	Set	One-to-one
Commissioned only	9/0/18	27/0/0	19/0/8	15/0/12
Original only	0/0/27	0/0/27	0/0/27	0/0/27
Union	9/0/18	27/0/0	20/0/7	16/1/10
Reciprocal	0/0/27	0/0/27	0/0/27	0/0/27

The commissioned-only one-to-one effect ranged from -0.0368 to 0.0572, with median 0.0175. The union effect ranged from -0.0368 to 0.0602, with median 0.0164. Original-only and reciprocal effects were negative throughout. Spatial support remained positive throughout commissioned-only and union training. These results separate ranker-free formation gains from ranker-dependent queued gains.

S6. Merge-aware bounds for the fixed U-Net queue

Set coverage is the number of distinct truth instances incident to at least one qualified selected candidate. One-to-one support is the maximum matching cardinality. Qualified memberships count all candidate-to-instance edges. Memberships beyond the set union equal total memberships minus distinct covered instances. The merge-capacity gap equals set-covered instances minus matched instances.

Table 10: Merge-aware accounting for the 518-region queue. Recall denominators contain 1,712 weed instances.

Subset	Regions	Zero-edge	Single	Multi	Set instances	1:1 instances
All selected	518	170	286	62	453	348
Consensus reviewable	301	52	195	54	346	249
Mask-qualified	355	7	286	62	453	348
Human and mask positive	249	0	195	54	346	249

Subset	Set recall	1:1 recall	Merge gap	Duplicate excess
All selected	0.2646	0.2033	105	0
Consensus reviewable	0.2021	0.1454	97	0
Mask-qualified	0.2646	0.2033	105	0
Human and mask positive	0.2021	0.1454	97	0

All 453 edge memberships represented distinct instances, so duplicate memberships beyond the set union were zero. The 105-instance all-queue gap was therefore attributable to merged capacity under the matching graph. A selected candidate qualified for at most nine weed instances.

The increase in reviewability with multiplicity shows that merged candidates can retain useful evidence. Set coverage and one-to-one support are computed on the same thresholded graph and are indexed by the queue capacity and matching tuple. The operator protocol did not request plant

Table 11: Operator consensus reviewability by qualified geometric multiplicity.

Qualified instances per region	Regions	Reviewable	Reviewable fraction	Unresolved
0	170	52	0.3059	39
1	286	195	0.6818	31
2	43	35	0.8140	4
3 or more	19	19	1.0000	0

counts; neither geometric endpoint is interpreted as realised human recovery or as a bound on it.

S7. Model stability, standard metrics, and crossed uncertainty

Table 12: Per-seed $K = 20$ queue results under the primary contract. Crop is any crop-exposed tile frequency.

Model	Seed	Precision	Spatial	Set	One-to-one	Crop
U-Net	20260810	0.7112	0.2704	0.2494	0.2062	0.2692
U-Net	20260811	0.7248	0.2634	0.2477	0.1992	0.3077
U-Net	20260812	0.6879	0.2523	0.2266	0.2033	0.2308
U-Net	20260813	0.7569	0.3008	0.2833	0.2097	0.2308
U-Net	20260814	0.7035	0.2471	0.2272	0.1951	0.1923
DeepLabV3+	20260820	0.6848	0.2763	0.2436	0.1945	0.2692
DeepLabV3+	20260821	0.6863	0.2588	0.2319	0.1887	0.2308
DeepLabV3+	20260822	0.6784	0.2780	0.2506	0.1939	0.2308
Mask R-CNN	20260820	0.8000	0.2453	0.2313	0.2202	0.2308
Mask R-CNN	20260821	0.7942	0.2371	0.2255	0.2132	0.1538
Mask R-CNN	20260822	0.7731	0.2418	0.2208	0.2144	0.3077

Table 13: Crossed seed-by-image contrasts for one-to-one support. Familywise intervals use Bonferroni-adjusted 97.5% percentiles for two prespecified contrasts.

Contrast	Difference	Crossed 95% interval	Familywise 97.5% interval
DeepLabV3+ minus U-Net	-0.0103	$[-0.0237,\ 0.0026]$	$[-0.0256,\ 0.0049]$
Mask R-CNN minus U-Net	$+0.0132$	$[-0.0009,\ 0.0300]$	$[-0.0027,\ 0.0329]$

Date-block plus seed intervals were $[-0.0244, 0.0133]$ for DeepLabV3+ minus U-Net and $[0.0023, 0.0341]$ for Mask R-CNN minus U-Net. They resample four observed dates and training seeds. They do not estimate a new-field distribution.

Table 14: Standard predicted-weed instance metrics. Each cell reports the seed mean with sample SD in parentheses.

Model	mAP	AP50	AP75	AR100	AR100@50
U-Net	0.0853 (0.0037)	0.2616 (0.0157)	0.0375 (0.0074)	0.1293 (0.0082)	0.3412 (0.0207)
DeepLabV3+	0.0499 (0.0026)	0.1937 (0.0060)	0.0093 (0.0034)	0.1010 (0.0054)	0.2950 (0.0081)
Mask R-CNN	0.1064 (0.0082)	0.3189 (0.0248)	0.0516 (0.0027)	0.1483 (0.0101)	0.3904 (0.0266)

Model	Queue AR20	Queue AR20@50	Queue PQ-style
U-Net	0.0699 (0.0020)	0.1672 (0.0070)	0.1773 (0.0055)
DeepLabV3+	0.0517 (0.0013)	0.1456 (0.0045)	0.1464 (0.0023)
Mask R-CNN	0.0782 (0.0049)	0.1861 (0.0079)	0.1936 (0.0085)

The standard calculations used at most 100 predicted weed regions per image. Queue AR20 used the predicted-weed subset of the $K = 20$ review queue. The PQ-style calculation used IoU greater than 0.50 and decomposed into segmentation and recognition terms. It is not a full panoptic score. Primary endpoint replay was checked for 66 model–seed–metric combinations; all differences were zero.

Across the 27 contracts, Mask R-CNN ranked first in 21 and U-Net in six. U-Net ranked first at instance coverage 0.75 with labelled coverage 0.25 or 0.50 across all three purity values. Mask R-CNN ranked first in all remaining cells. DeepLabV3+ did not rank first.

S8. Operator agreement and human–mask correspondence

The two operators were different natural persons. They followed common written instructions and completed 12 validation-derived practice items. They independently reviewed every formal item without discussion, peer-response access, automatic labelling, or external search. No occupational, demographic, or professional credential data were collected. Results are therefore restricted to these two anonymous operators. Timing fields were excluded because timing was not part of the protocol.

Table 15: Agreement on 518 regions. AC1 intervals use 10,000 scene-cluster replicates.

Field	Exact agreement	Cohen's κ	Gwet's AC1	AC1 95% interval
Primary content	0.9479	0.9095	0.9355	[0.9095, 0.9593]
Weed reviewability	0.9421	0.8930	0.9206	[0.8918, 0.9483]
Crop presence	0.9459	0.8492	0.9341	[0.9081, 0.9579]
Localisation usability	0.9402	0.9049	0.9243	[0.8998, 0.9481]
Confidence	0.9440	0.9160	0.9328	[0.9100, 0.9540]

For 445 determinate localisation pairs, exact agreement was 0.9955, linear weighted κ was 0.9944, and quadratic weighted κ was 0.9966. For confidence, linear and quadratic weighted κ were 0.9366 and 0.9589. Mean confidence was 3.523 for operator A and 3.510 for operator B. Both medians were 4 and both interquartile ranges were 3–4.

Table 16: Weed-reviewability contingency table. Rows are operator A and columns are operator B.

	No	Yes	Uncertain
No	143	0	3
Yes	0	301	16
Uncertain	1	10	44

Table 17: Human consensus versus mask qualification for all 518 regions. Unresolved includes disagreement or at least one uncertain response and is excluded from binary correspondence metrics.

Human status	Mask positive	Mask negative
Consensus reviewable	249	52
Consensus not reviewable	67	76
Unresolved	39	35

Binary correspondence was evaluated on the 444 determinate-consensus regions: true positives = 249, false positives = 67, false negatives = 52, and true negatives = 76. The remaining 74 regions stayed in the three-by-two contingency table but did not enter binary denominators. Mask precision against human consensus was 0.7880, recall was 0.8272, F1 was 0.8071, Jaccard was 0.6766, and accuracy was 0.7320. The determinate analysis covered 85.71% of all reviewed regions.

Table 18: Five-point confidence contingency table. Rows are operator A and columns are operator B.

	1	2	3	4	5
1	0	0	0	0	0
2	0	44	11	0	0
3	0	18	172	0	0
4	0	0	0	220	0
5	0	0	0	0	53

S9. Pre-specified CWFID transfer and continuous diagnostics

The transfer pipeline scaled each longer image side to 600 pixels, reflection-padded the shorter side, and applied training normalisation. Annotations used nearest-neighbour resizing and zero padding. Five U-Net checkpoints retained their validation-selected epoch, foreground threshold, minimum area, and ranking score. CWFID labels were not used for fitting, calibration, threshold selection, or allocation.

Table 19: Pre-specified CWFID endpoint. Every checkpoint produced the same structural outcome.

Check-points	Images	Weed components	Candidates	Accepted	One-to-one	Crop exposure
5	60	331	0	0	0/331	0/60

The all-zero empirical image bootstrap is degenerate and is not used as a population interval. The one-sided 95% binomial reference upper bound for candidate presence in an image was 0.0487. This quantity does not bound weed-component recall or transfer to other farms.

Twenty equal-width calibration bins were specified. Almost all pixels fell below predicted plant probability 0.05. In that bin, mean predicted probability ranged from 0.0072 to 0.0239 across checkpoints, while observed plant prevalence was about 0.055. Higher-probability bins were sparse and often had near-zero observed plant fraction.

Table 20: Post-lock continuous CWFID diagnostics. AUROC and AP are image-macro plant-pixel scores.

Checkpoint	Validation threshold	Plant AUROC	Plant AP	Foreground $q_{0.99}$	Foreground maximum	Images with candidates
1	0.50	0.3871	0.0438	0.0131	0.0184	0
2	0.70	0.4931	0.0509	0.0407	0.2205	0
3	0.30	0.6374	0.0715	0.0198	0.0613	0
4	0.50	0.5514	0.0577	0.0498	0.2122	0
5	0.50	0.5979	0.0677	0.0118	0.0384	0

The diagnostic threshold grid was 0.01, 0.02, 0.05, 0.10, 0.20, 0.30, 0.40, 0.50, 0.60, 0.70, 0.80, 0.90, and 0.95. At 0.01, spatial recall ranged from 0.0363 to 1.0000. Only checkpoint 2 obtained nonzero role-qualified recall, at 0.0423. Several lower-threshold components were broad or mixed. The diagnostic therefore cannot distinguish threshold mismatch from representation, role-semantic, scale, crop, viewpoint, or annotation-contract shift.

External evaluation used the five U-Net checkpoints available in the pre-specified transfer. The later Mask R-CNN models were evaluated on WeedsGalore and provide a defined model family for future cross-dataset comparison.

S10. Candidate-ranking and allocation diagnostics

Table 21: Candidate discrimination on the complete 2,351-candidate WeedsGalore test pool. AUC and AP are diagnostics.

Ranker and fitting support	AUC	AP
Source weed score	0.4947	0.2877
Role plus local, all candidates	0.9049	0.6960
Geometry, all candidates	0.8154	0.5418
Frozen DINOv2, all candidates	0.9630	0.8684
Component-masked DINOv2, all candidates	0.9665	0.8813

The masked DINOv2 ranker reached one-to-one support 0.1343 at $K = 20$ despite its candidate AUC of 0.9665. This gap motivates separate candidate and instance-support reporting.

An exploratory 5–50 candidates-per-tile allocation was evaluated after official-test access. It was not selected by validation in the five-seed study.

Fixed $K = 20$, bounded allocation, and global top-N accepted 520 regions. Their one-to-one numerators were 227, 315, and 320. The bounded and global policies concentrated more than half of reviews on 6 June and reduced worst-date support. These results remain hypothesis-generating.

S11. Metric correspondence

Table 22: Relationship between established instance metrics and the finite-queue contract.

Metric	Matching or aggregation	Capacity representation	Information emphasised
mAP	score-ranked one-to-one matches across IoU thresholds	maximum detections configured externally	precision–recall and mask localisation
AR	recall across IoU thresholds and detection limits	explicit detection limit, here 100 or 20	localisation recall under a count cap
PQ	one-to-one IoU matching with recognition and segmentation terms	no agricultural review budget	complete panoptic partition quality
Beyond-mAP diagnostics	explicit duplicate and category-error analysis	not tied to a review queue	duplicate and semantic error modes
SortedAP	matching over exact IoU change points	not tied to an agricultural budget	sensitivity and continuity of instance quality
Set coverage	union of all qualified candidate–instance edges	fixed reviewed-region budget	merge-aware geometric support
One-to-one queue support	maximum qualified matching	fixed per-tile review budget	distinct one-region-to-one-record support

The queue endpoint reuses familiar matching machinery. Its contribution is the agricultural information contract: it records proposal support, role qualification, merge capacity, ranker source, and review truncation as separate evidence layers. It complements AP, AR, PQ, and recent instance-quality metrics with a fixed review-capacity endpoint.

S12. Reproducibility resources

The release contains versioned configurations, model-acquisition records, environment specifications, training and evaluation source, checkpoint hashes, aggregate and per-image metrics, matching outputs, and figure source data. Public dataset imagery and weights are acquired from upstream sources. They are not redistributed when upstream terms prohibit redistribution.

The principal environment used Python 3.10.12, PyTorch 2.7.1 with CUDA 12.6, torchvision 0.22.1, and segmentation-models-pytorch 0.5.0. Training used one NVIDIA H200 accelerator per seed. Physical devices 6 and 7 were explicitly assigned. CPU-only aggregation and bootstrap programs read immutable result tables.

The compact verifier reconstructs published aggregate tables and numerical invariants from archived outputs and verifies their checksums. Full reproduction uses the public datasets and weights, recorded environment, and supplied training and evaluation programs. The public package contains only publication-ready aggregate operator results and excludes private response files.